

Daniel Mandragona

ML Performance Engineer | TPU kernels, parallelism & quantum information

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EXPERIENCE

Software Engineer, ML Performance Engineering (MaxText)

Apr 2026 – Present

Google

Mountain View, CA

- Delivered a 40% speedup on a TPU Fast Fourier Transform (FFT) kernel for a customer engagement.
- Drove a 2× speedup on a state-of-the-art model for a strategic, high-priority customer via kernel fusions, parallelism strategies, and per-kernel tuning.
- Microbenchmarked the Splash Attention kernel, isolating bottlenecks to extract a further 10% speedup on an already heavily optimized kernel.
- Performance experimentation on the Wan 2.1 diffusion model, achieving a 25% speedup via Unified Sequence Parallelism (USP) — a hybrid long-context scheme combining Ring Attention and DeepSpeed-Ulysses sequence parallelism.
- Roofline analysis of Qwen 3.5 against next-generation TPU specifications, projecting achievable utilization and isolating compute- vs. memory-bound regimes to inform hardware/software co-design.
- Identified, researched, implemented, and benchmarked an mHC-lite optimization for DeepSeek V4 that sidesteps the costly iterative Sinkhorn algorithm used to generate doubly-stochastic expansion matrices.

20% Research Engineer, Device Modeling & Optimization

Mar 2025 – Sep 2025

Google

Mountain View, CA

- Signal processing and statistical modeling (e.g., peak-finding, Skew-Normal/GMM fitting) to identify and characterize Two-Level System (TLS) features in noisy qubit data.
- Researched adaptive TLS thresholding and created custom metrics to assess algorithm performance.
- Computer vision techniques for identifying stationary TLS features in heatmap data.
- Refined unsupervised learning algorithms through rigorous data analysis, employing statistical summaries and visualizations to enhance high-level feature understanding.

Software Engineer, Certificate Authority Infrastructure

Sep 2024 – Apr 2026 & Aug 2019 – 2022

Google

Mountain View, CA

- Optimized Certificate Authority performance by diagnosing and resolving complex low-level issues, including Go/C interop bugs, database hotspot thrashing, network stack limitations (e.g., network egress), and DNS resolver latency.
- Designed and prototyped novel load balancing solutions for unique networking constraints, and implemented server migrations into highly sensitive & secure machine pools.
- Full-Lifecycle software development including implementing RFC security protocols (e.g., ACME RFC8555), building performance/integration testing frameworks, and ensuring system robustness for public release. Discovered and [reported](#) a critical DoS vulnerability in Go/Crypto's DSA implementation.
- Held on-call responsibilities, managed rollouts, triaged outages, and contributed to datacenter compliance [WebTrust](#) audits and Key Destruction Ceremonies.

Software Engineer, Engineering Residency

Sep 2018 – Aug 2019

Google

Mountain View, CA

- **Google Research — Perception:** Developed and optimized a large-scale ML pipeline, utilizing MapReduce to process billions of data points for Image Saliency Prediction. Performed hyperparameter tuning on a Resnet Unet model architecture, and implemented state-of-the-art metric functions for evaluation.
- **Android — Play Abuse:** Built multimodal ML infrastructure for Abuse Detection in the Google Play Store, decoupling feature generation from model inference.

Research Collaborator

FermiLab

May 2024 – Oct 2024
College Station, TX

- Researched quantum error correcting spherical codes for qudits, simulating them in Python using QuTiP, and benchmarking their error performance across various noise models.

Teaching Assistant

Texas A&M's Department of Mathematics

Aug 2022 – Aug 2024
College Station, TX

Teaching Assistant for CS1

UCF's Department of Computer Science

Aug 2017 – May 2018
Orlando, FL

EDUCATION

Masters in Mathematics, GPA: 3.93

Texas A&M University

Aug 2022 – Dec 2024
College Station, TX

- **Qualifying Exams Passed:** Real Analysis and Topology/Differential Geometry.

BS Mathematics; Concentration in Computational Sciences, GPA: 3.71

University of Central Florida

Aug 2013 – May 2018
Orlando, FL

- Honors in the Major.

Quantum Information Science Summer School

Oak Ridge National Laboratory, TN

July 2024

Focus on topological quantum computation, and quantum software tools.

TECHNICAL SKILLS

Languages & Tools

Python · C++ · Golang · Qiskit · MATLAB · C · Java · Mathematica · SQL

Coursework

Quantum Algorithms · Spectral Theory for Schrödinger Operators · Real Analysis Sequence · Probability Theory · Physics for Mathematicians · Functional Analysis · Differential Geometry Sequence · Algebra Sequence

PRESENTATIONS

QEC: From Classical Errors to the Surface Code — Quantum AI, Google

Oct 2025

Presented fundamental classical and quantum error correction theory, starting with the Hamming Code and building up to the surface code for QEC.

Berry Phase & Chern Numbers — Master's Presentation, TAMU

Nov 2024

Mathematical foundations of Berry phase and Chern numbers, including their gauge invariance, discrete and continuous formulations, and applications in condensed matter physics. Implemented numerical methods to compute Chern numbers for topological systems, reproducing results from published research.

Quantum Markov Chains — Quantum Algorithms, CSCE 640 – TAMU

Nov 2023

Presented basic theory of Markov chains, and how the proposed Quantization scheme outlined by Szegedy leads mathematically to a quadratic speedup in convergence to the stationary distribution over the classical version. Includes a further description of the quantization of the Monte Carlo Metropolis-Hastings algorithm.

Weyl Quantization Lecture — Topics in Physics for Mathematicians, MATH 689 – TAMU

Dec 2023

Presented the mathematical theory for converting classical phase-space $L^2(\mathbb{R}^{2n})$ -observables to be self-adjoint operators on a quantum Hilbert space.

Visual Saliency Prediction — Perception Research Showcase, Google

Mar 2019

Presented the topic of image saliency and its motivations, and the ML infrastructure my team used for prediction and evaluation.

Functional Programming — Engineering Residency Program, Google

Oct 2018

Taught Engineering Residency cohort about functional programming fundamentals such as functors and monads in the context of the Haskell programming language.

Hopf Bifurcation Analysis — Undergraduate Research Excellence Showcase, UCF

Apr 2018

Conducted Hopf bifurcation research in a system of ODEs arising from a chemical reactor model. Utilized Mathematica software to perform the necessary symbolic computations for this analysis.

PUBLICATIONS

A Chaotic Chemical Reactor With and Without Delay: Bifurcations, Competitive Modes, and Amplitude Death.

S. Roy Choudhury and Daniel Mandragona. *Int. J. Bifurc. Chaos*, 2019.